

Security in IP-Based IoT Node and Device Authentication

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Abstract- The IoT security aims for enabling IoT data protection in various interconnected nodes. These frameworks require secured solutions to prohibit evaporation of personal information and harmful activate ventures by means of peer authentication and secure data transmission between the IoT nodes and devices. Node communication in IoT network sometime get hack and wrong operation performs if it happened then there will be a chance to go for heavy loss. So that proposed methodology improves the communication network with the implementation of IP binding technique and the uses of random encryption selection process. As there are the different nodes like receiver and sender data will be send to receiver in which the receive data need to be valid and decrypted using shared keys into the packet. At the receiver end before transferring the data to the IoT node it will be perform with IP extraction packet with IP validation on to the received packet which leads to perform secure data sharing at the both receiver and sender node. As per the above description the proposed mechanism will help for maintaining the security in IoT network.

Index Terms- Arduino, IoT, IoT Security, Network Security, NodeMCU, IoT Protocol, IP -Binding, Encryption, Decryption, IoT Network.

I. INTRODUCTION

The IP based IoT devices enables various smart electronics devices. These electronic devices are equipped with sensing, acting, and wireless communication capabilities. These capabilities are beneficial to interact and cooperate with each other in a universal way by means of IP connectivity. A Smart device is used to connect with IP based IoT node and devices as well as can be easily operate [1].

Each one electronics devices getting to be controlled and can we operate through the radiotelephone at 2050 .So,for every second data will be stored and retrieved.IoT architecture mostly consist of sensors which are connected to IoT board so,this particular IoT board is called as Node.The numbers of Node will be connected into single network which will be called as IoT Networks.Whole data will be passed through this IoT Networks so,may their will be a chance to the inturder to hack this networks which may cause a big loss.So,by implimenting Internet of things Secure Protocol System it is possible to avoid this loss [2,3].

To improve this sender-receiver verification technique is get propose. Here the data is get transfer to the user with highly encryption and key transfer technique so that the data in network is remain secured in this the system will also check for the sender side and receiver side.

If the side is right the cross verification is done otherwise not [4, 5].secure communication channel is established by IoT nodes and devices; while their connection should be bootstrapped through the process of node and device binding .In client server network of applications, client can be to verify and identity server with third party, binding does not collusion to bootstrap the communication between the client applications and the earmarked process of binding of IP based IoT node and device as shown in figure 1.

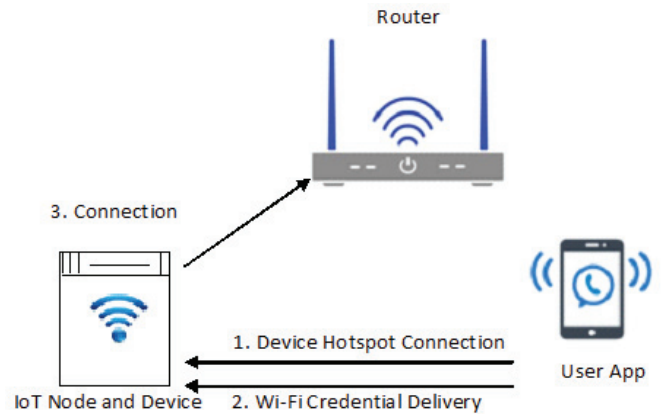


Fig. 1. Establish the connection between Node and Device

In figure 1, each device has its device password, in order that single authenticated user perceivable the password and connect with the device this procedure is called as IP based manual identification. Meanwhile the user connects with the device; the client will select their Wi-Fi connection for the device to connect. Finally the device and application will find each other on the Wi-Fi network [6]. Firstly the users need to connect with the specific device according to its IP address, the authorisation of Wi-Fi will be delivered to that device only. For better understanding the problem we can get the first step to represent proper study on IoT node and device binding mechanisms along with the secure connections [7, 8].In second step it is manually check the IoT node and device authentication process and found that third party attacks can be genuine implemented in IP based IoT devices[9,10].This research discloses the major failure that the doubt of the restricted environment, which frequently causes the absence of authentication throughout the process of node and device binding. Ultimately, this permit close intruders, such as neighbours, secret dispatch; third party attacks and to completely handle the sacrifice's device [11].

II. METHODOLOGY

Main aim to authenticate IoT Nodes and Devices based on unique IP address. Mostly IoT devices exchange their traffic regularly with servers. If we know these servers; we are able to identify IoT devices by observing traffic at during packet exchanges procedure also identify the types of nodes and their devices. Major goal is to find out virtual node creator in network as well as virtual device creator in network based on unique IP address.

Extract the IP of IoT device in a given node bind with the signal and send to receiver. Perform encryption on the signal data or node data of the IP. At the receiver end data is going to be received and decrypted. Then splitting of IP address and signal will be workout.

Check the retrieved IP address by trust list bind with receiver. If it's found then data will be in else it's going to be crosscheck with the other receiver trust list. On the basis of that IP bound factor will be calculated. Depend on that the authenticity of the IP will be measure and decision will be taken for acceptance or rejection.

A. IP Binding Model

In this technique the binding of IP address getting to be drill. Here the IP of sender and receiver is get bind with packet of message and send over the network. To which the IP address also get encrypted if any message is get altered then bind IP of the sender is also get updated. Now the receiver side will able to identify the exact sender IP address in which the sender can be easily identify by pattern matching implementation over the IP of sender and receiver. If the message get altered then receiver comes to know that the data signal get alter by intruder and will not perform the operation send the notification to admin IP binding sender IP and Receiver IP cross verification plays role of maintain security over the data signal transmission so that false message will not get delivered to the IoT boards which only restrict to the right sender and receiver terminal.

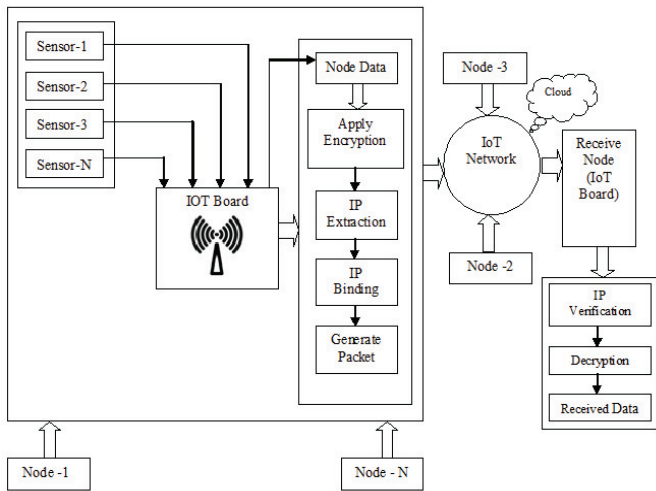


Fig. 2. Advance Architecture of Secure Protocol System for IoT

Advance Architecture of proposed secure protocol system is represented in figure 2. In which there is receiver node and sender node which are connected in IoT network. In which sender node will receive the data from IoT board and perform IP binding, then encryption over the received data and form a signal packet of that which is latterly shared

to the cloud via IoT network. At the other end signal is going to be received which latterly proceed with IP verification and decryption, which delivered the actual data to the receiver node according to the decision are taken by the receiver.

III. DESIGN FOR SECURE COMMUNICATION

As per the existing study in IoT Networks most probably data will be shared without any authentication and verification which cause big security clause in IoT Networks.

So, it is necessary to improve data sharing with some advance techniques. So, that we propose a mechanism which authenticate and verify data at the time of sending as well as receiving which make the IoT network more secure. Through figure 2 this window performs Device as well as Node login as shown in figure 3. Every Node is configured with Username and Password. The Particular Node is going to be allotted with Username and Password. That Data will be considered for Authentication of credentials. A login form is used to enter authentication credentials to access a restricted form. The login form contains a field for the username and password. When the login form was submitted its underlying code checks that the credentials are authentic or not [12].



Fig. 3. Node and Device Login Panel

A. Virtual Node Creator in Network

In figure 4 first to create a node and add number of devices for every device we are going to manipulate data as per the algorithm given there are various range of algorithm for encrypting the data. So, every node we receive and send the data using different encryption algorithms, for authentication, integration, confidentiality and encryption from different devices used on IoT [13,14].

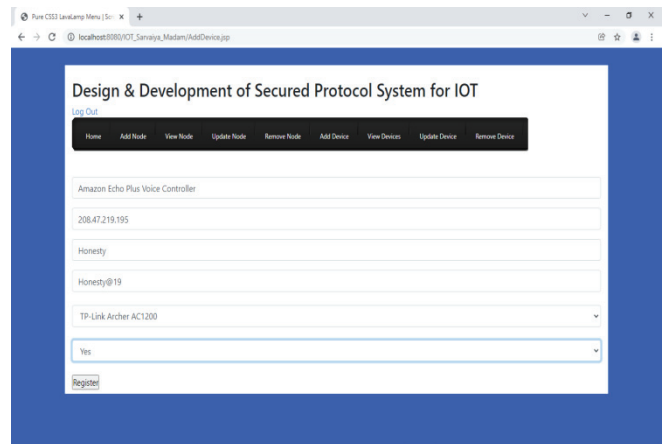


Fig. 4. Virtual Node Creator in Network

B. Virtual Device Creator in Network

In figure 5 first register the required IoT devices with IP Address along with username, password and IoT node name. Understand the risk of unsecure IoT nodes and devices requires location information with respect to; latitudinal and longitudinal way then registered IP-based IoT devices [15].

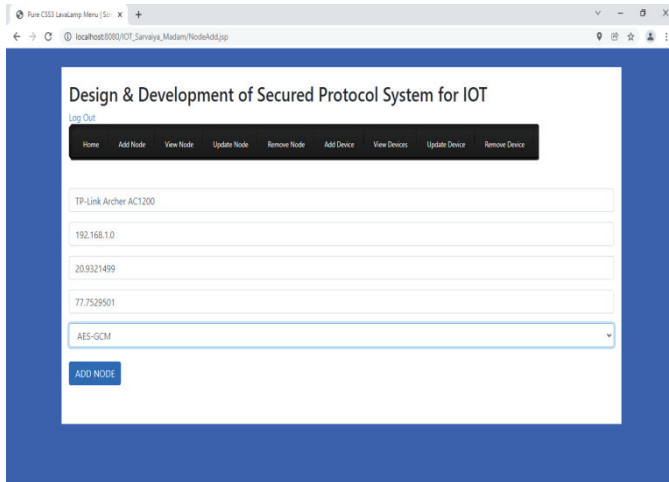


Fig. 5. Virtual Device Creator in Network

IV. SECURITY ANALYSIS

The Dashboard in figure 6 represents the list of Nodes and it's connected Devices. This Dashboard also helps to control all Nodes transmission and their data. Secure communication is required since the adversary can eavesdrop on traffic in the IoT network and maliciously modify it. Two communicating parties must establish a pairwise key to assure authentication [16].

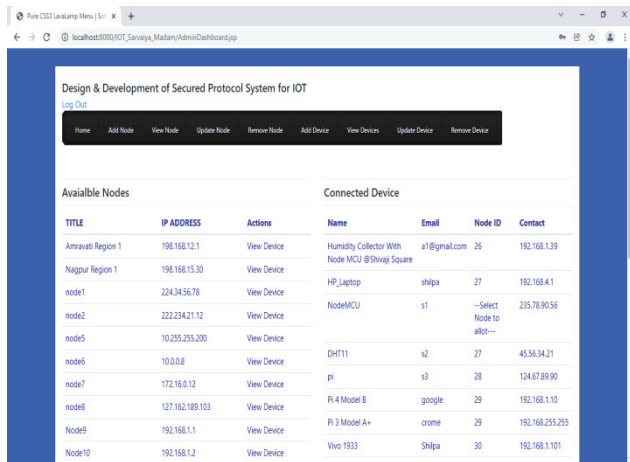


Fig. 6. Admin Dashboard with Node and Device Details

V. SECURE COMMUNICATION SYSTEM'S FLOW CHART

The goal of the propose model is to build a secure protocol system for the IoT network to decrease time, memory consumption. Also provide suitable security mechanisms for the IoT security layers. The proposed model helps researchers and designers to select the convenient protocols and security mechanisms for each security layer to secure data and smart objects. In the proposed model, we use Things Board platform which provides many security mechanisms. Also, manage the strategy of selection security

algorithms to achieve high level of security requirements and decrease power consumption and time [17, 18].

The graphical representation of proposed secure protocol system is represented in figure 7.

The procedure of the proposed model work flow chart can be presented in Figure 7 as follows:

- Collecting to nodes and bind with the signal and send to NodeMCU ESP8266.
- Message or signal to send.
- Add sender IP to transmitted data.
- Perform encryption on data and sender IP.
- Received data over the node.
- Decryption of receiver IP and signal.
- Verify IP of receiver.
- If IP match with IP list then and only then allow data receiving.
- Else it's going to be cross check with the other receiver trust list.
- Data will be found then stop the process.

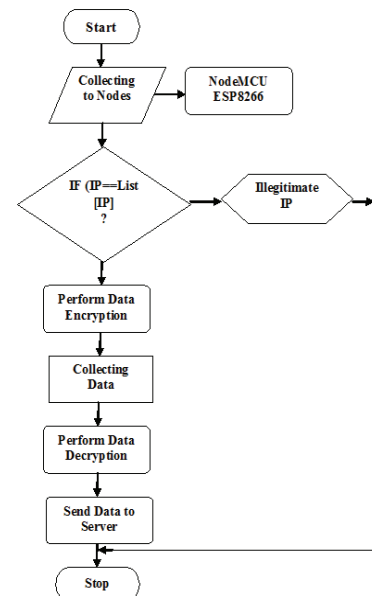


Fig. 7. Proposed Secure System Flow Chart

The following flowchart shows the step by step approach to solving a task:

Step 1: Exact the IP of IoT device in a given node bind with the signal and send to receiver.

Step 2: Perform Encryption on the signal data or node data of the IP.

Step 3: The receiver side data is going to be received and decrypted. Then splitting of IP address and signal will be workout.

Step 4: Checking of the retrieved IP address with trust list bind with receiver.

Step 5: If it's found then data will be in else it's going to be crosscheck with the other receiver trust list.

Step 6: On the basis of that IP bound factor will be calculated.

Step 7: Depend on that the authenticity of the IP will be measure and decision will be taken for acceptance or rejection.

Our proposed algorithm is Secure Protocol System for IoT (**SPSIOT**) workout in the following steps:

STEPS:

- ```

1. Start()
2. Bindedip=null;
3. IP=GetIP(Extract_ip_sender);
4. splited_digits_ip=Split_Ip_Data(IP);
5. for single_digits insplited_digits_ip;
6. char_digit=ConvertToChar(single_digit);
7. ASCII_digit=GetAsciiValue(char_digit);
8. rand= GetRand();
9. final_digit=ASCII_digit+Rand;
10. final_digit=final_digit+Tail_Factor();
11. form_encoded_ip=ApplyEncoding(final_digit);
12. Output(form_encoded_ip);
13. Finish;

```

**Explanation:**

Step 1: First of all we start the function.

Start ()

Step 2: Determine the variable Bindeip is equal to null that is

```

Bindeip = null;

```

Step 3: Extract the IP by using the function GetIP ( ) with Extract\_ip\_sender parameter.

```
GetIP (Extract ip sender);
```

Step 4: After getting IP split the IP by using `split_Ip_Data()` with parameter IP and transfer the value to `spilted_digits ip` variable.

```
splited_digits_ip = split Ip Data(IP);
```

```

Step 5: After that we get single_digit from
splited_digits_ip. single_digits = splited-
digits-ip;

```

Step 6: Convert the single digit to character digit by using `ConvertToChar ( )` with parameter `single digit`.

```
char_digit = ConvertToChar
 (single digit);
```

Step 7: Then convert the single digits to ASCII value.

```
ASCII digit = ConvertToASCII(char digit);
```

Step 8: Apply randomization function with the help of GetRand ().

```
rand = GetRand ();
```

Step 9: Adding ASCII\_digit with randomization function we get final digit.

```
final digit = ASCII digit + Rand;
```

Step 10: Final out final digit after adding final digit  
with tail factor function                      final digit =

```
final_digit + Tail_Factor();
```

Step 11: Form encoded IP by applying encoding to final digit.                      form encoded ip =

ApplyEncoding (final-digit);  
Step 12: Form encoded IP as an output that is

```
IP->Char->INT->ASCII->rand-
>form encoded ip
```

Step 13: Final IP is going to be transfer as a sender side this same IP is also to be decrypted at the other end that is receiver side.

Step 14: Over that perform decryption again split that into reverse order, get the number  
Perform your encoded IP.

## VII. RESULT AND ANALYSIS

As per the propose system the practical evaluation shown into the figure 8. In which the first window will shown implementation and result of various nodes are connected to the IoT network .To which the parameters like trust factor, encrypted IP, encrypted receiver message is shown.

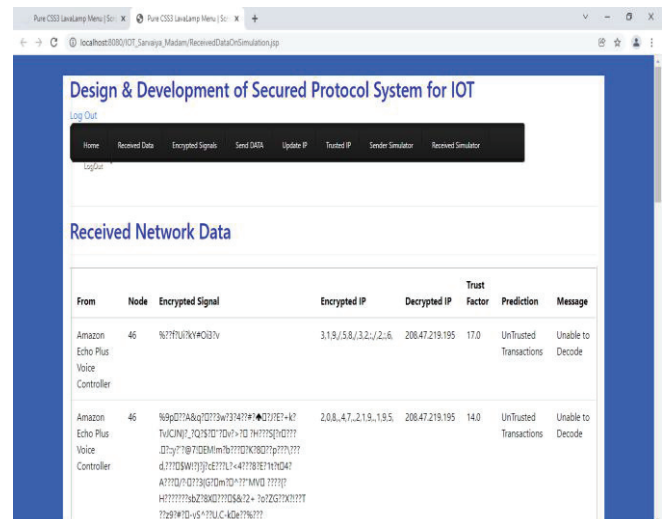


Fig. 8. Received Network Data

In Figure 9 terminal output of NodeMCU shows the device data collected and send to the server or IoT network. As there are the different parameters shown the main parameter like encrypted message, encrypted IP used for result evaluation. It is seen to be that the proposed network protocol helps to increase the security of IoT network.

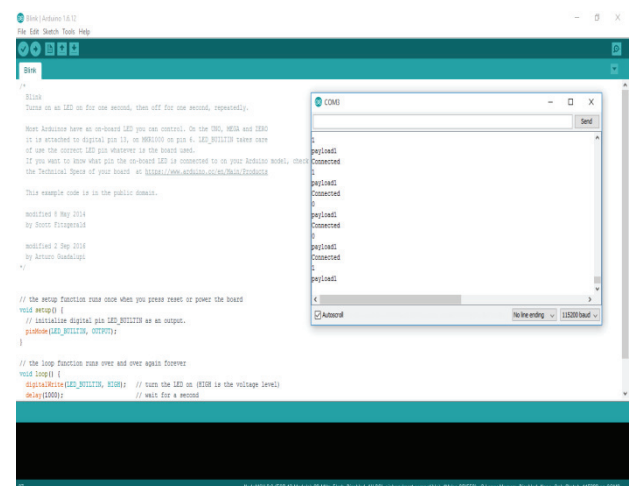


Fig. 9. Active Object Detection

## VIII. CONCLUSION AND FUTURE WORK

As per the existing study in IoT Networks most probably data will be shared without any authentication and verification which cause big security clause in IoT Networks. So, it is necessary to improve data sharing with some advance techniques. So that the propose mechanism which authenticate and verify data at the time of sending as well as receiving which make the IoT network more secure.

If there is any attacks perform in IoT network then due to the authentication the intruder can be trap and system will not allowed to change the network data. The goal of this module helps researchers and designers to select the convenient protocols and security mechanisms for each security layer to secure data and smart objects. The secure protocol system is use to prevent or decrease attacks, threats and various types of problems that occur during the device authentication process.

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